Problem Statement: Financial Transaction ProcessingSystem (Diamond Inheritance)

Design and implement a C++ program for a Financial Transaction Processing System to demonstrate the concept of Diamond Inheritance.
In a financial organization, different departments handle operations related to a customer's account. Some departments deal with deposit operations, while others manage loan services. Both departments require access to common account information. A central financial service module combines both functionalities to process complete financial transactions.
This scenario forms a Diamond Inheritance structure, which should be handled using virtual inheritance to avoid ambiguity.

Requirements
1. Create Base Class Account
Data Members
• accountNumber
• accountHolderName
Member Functions
• acceptAccountDetails() – Accept account information from the user
• displayAccountDetails() – Display account information

2. Create Derived Class DepositService
Inherit virtually from class Account.
Additional Data Member
• depositAmount
Member Functions
• acceptDeposit() – Accept deposit amount
• displayDeposit() – Display deposit details

3. Create Derived Class LoanService
Inherit virtually from class Account.
Additional Data Member
• loanAmount
Member Functions
• acceptLoan() – Accept loan amount
• displayLoan() – Display loan details

4. Create Derived Class FinancialService
This class inherits from both:

• DepositService
• LoanService
Member Function
• displayFinancialSummary() – Display complete financial information including account details, deposit amount, and loan amount.

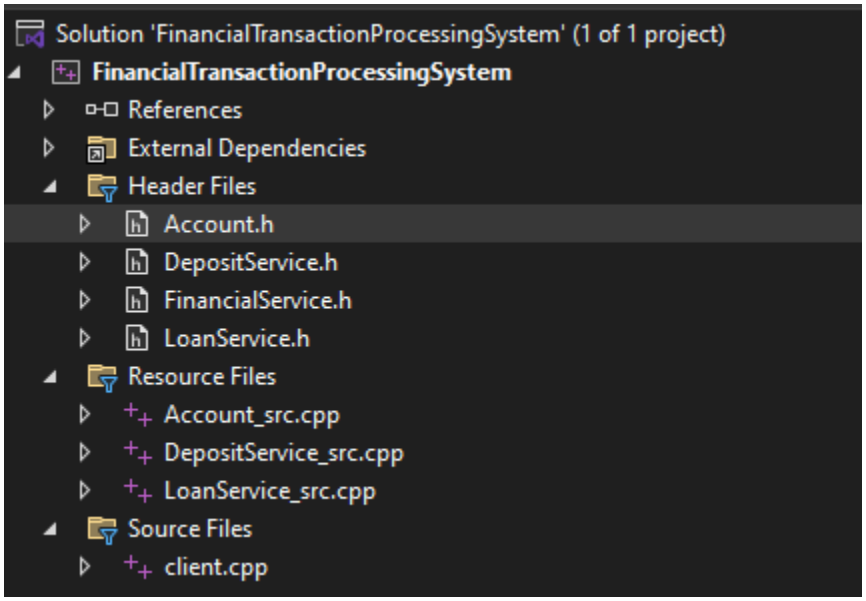
5. In main() Function
• Create an object of class FinancialService.
• Accept account details, deposit amount, and loan amount.
• Display the complete financial summary.

Objective of the Program
The program should demonstrate:
• Diamond Inheritance in C++
• Multiple inheritance from a common base class
• Ambiguity problem when accessing base class members
• Resolution using virtual inheritance
• Accessing common account information from the final derived class.

Expected Class Hierarchy
Account
/ \
/ \
DepositService LoanService
\ /
\ /
FinancialService

Solution

Structure:



File Location:

Header File

./Account.h

```
#pragma once
#include<iostream>
using namespace std;

class Account {
private:
	int accountNumber;
	string accountHolderName;
public:
	Account();
	Account(int accountNumber, string accountHolderName);
	void acceptAccountDetail();
	void displayAccountDetail();
};
```

./DepositService

```
#pragma once
#include"Account.h"

class DepositService :public Account {
private:
	int depositAmount;
public:
	DepositService();
	DepositService( int depositAmount);
	void acceptDeposit();
	void displayDeposit();

};
```

./LoanService.h

```
#pragma once
#include"Account.h"

class LoanService :virtual public Account {
private:
	int loanAmount;
public:
	LoanService();
	LoanService(int loanAmount);
	void accpetLoan();
	void displayLoan();
};
```

./FinancialService.h

```
#pragma once
#include"DepositService.h"
#include"LoanService.h"

class FinancialService :public DepositService, public LoanService {
public:
	void acceptFinancialDetail() {
		Account::acceptAccountDetail();
		DepositService::acceptDeposit();
		LoanService::accpetLoan();
	}
	void displayFinancialSummary() {
		Account::displayAccountDetail();
		DepositService::displayDeposit();
		LoanService::displayLoan();
	}
};
```

Resource Files

./Account_src.cpp

```
#include"Account.h"

Account::Account(){
	accountNumber = 0;
	accountHolderName = "Unknown";
}
Account::Account(int accountNumber, string accountHolderName) {
	this->accountNumber = accountNumber;
	this->accountHolderName = accountHolderName;
}

void Account::acceptAccountDetail() {
	cout << "\nEnter Account Details---\n\n";
	cout << "\nEnter Account Number : ";
	cin >> accountNumber;
	cout << "\nEnter Account Holder Name : ";
	cin >> accountHolderName;
}

void Account::displayAccountDetail() {
	cout << "\nAccount Details : \n";
	cout << "\nAccount Number : " << accountNumber;
	cout << "\nAccount Holder Name : " << accountHolderName;
}
```

./DepositService_src.cpp

```
#include"DepositService.h"

DepositService::DepositService() {
	int depositAmount = 0;
}

DepositService::DepositService(int depositAmount)
{
	this->depositAmount = depositAmount;
}

void DepositService::acceptDeposit() {
	cout << "\nEnter Amount to Deposit : ";
	cin >> depositAmount;
}
void DepositService::displayDeposit() {
	cout << "\nDeposited Amount : " << depositAmount;
}
```

./LoanService

```
#include"LoanService.h"

LoanService::LoanService() {
	int loanAmount = 0;
}

LoanService::LoanService (int loanAmount)
{	this->loanAmount = loanAmount;
}

void LoanService::accpetLoan() {
	cout << "\nEnter Loan Amount : ";
	cin >> loanAmount;
}
void LoanService::displayLoan() {
	cout << "\nLoan Amount : " << loanAmount;
}
```

Source/Client File

./client.cpp

```
#include"FinancialService.h"

int main() {

	FinancialService f1;
	f1.acceptFinancialDetail();
	f1.displayFinancialSummary();
}
```

Output

By Using Dynamic Array

./Client.cpp

```
#include"FinancialService.h"

int main() {
	int noe;

	cout << "Enter No of Customer accounts : ";
	cin >> noe;
	FinancialService* faa=new FinancialService[noe];

	for (int i = 0; i < noe; ++i) {
		cout << "\n--- Input Details for Account " << i + 1 << " ---";
		faa[i].acceptFinancialDetails();
	}

	for (int i = 0; i < noe; i++) {
		cout << "\n--- Summary for Account " << i + 1 << " ---";
		faa[i].displayFinancialSummary();
	}

	delete[]faa;
	faa = NULL;
	return 0;

}
```

Output

```
Microsoft Visual Studio Debug Console
Enter No of Customer accounts : 3

--- Input Details for Account 1 ---
Enter Account Details-----

Enter Account Number : 963852

Enter Account Holder Name : Vikas

Enter Amount to Deposit : 15000

Enter Loan Amount : 5000

--- Input Details for Account 2 ---
Enter Account Details-----

Enter Account Number : 741852

Enter Account Holder Name : Ashutosh

Enter Amount to Deposit : 45600

Enter Loan Amount : 32100

--- Input Details for Account 3 ---
Enter Account Details-----

Enter Account Number : 357951

Enter Account Holder Name : Vikas

Enter Amount to Deposit : 350499

Enter Loan Amount : 150000
```

```
--- Summary for Account 1 ---
Account details are:
Account Number : 963852
Account Holder Name : Vikas
Deposited Amount : 15000
Loan Amount : 5000
--- Summary for Account 2 ---
Account details are:
Account Number : 741852
Account Holder Name : Ashutosh
Deposited Amount : 45600
Loan Amount : 32100
--- Summary for Account 3 ---
Account details are:
Account Number : 357951
Account Holder Name : Vikas
Deposited Amount : 350499
Loan Amount : 150000
D:\LearnC++\Day7_05032026\FinancialTransactionPr
ss 7104) exited with code 0 (0x0).
Press any key to close this window . . .
```